NGUYEN VAN DAT

ROBOTICS R&D ENGINEER

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github.com/blueDstar | linkedin.com/in/vandat152 | Portfolio

Robotics and Artificial Intelligence graduate focused on robotic systems, control, embedded platforms and computer vision. Hands-on experience in SolidWorks CAD, mechanism simulation, CAN-bus control, ROS2 and edge-AI deployment.

01 PROFESSIONAL EXPERIENCE

LYK / Hilykan

08/2026 - 10/2026

Mechatronics & Robotic Arm R&D Engineer

- Researched Damiao DM-J4310 actuators for OpenArm: hardware connections, simulation and CAN-bus control experiments.
- Created SolidWorks 3D models, assemblies and technical drawings for a smartwatch fixture, vision station and Shoe QC Box.
- Simulated vision-station and fixture/clamping movements against operating requirements.

DAT Group

01/2026 - 04/2026

Research & Development Intern

- Developed Vision AI for human recognition, abnormal-event and foreign-object detection; deployed on Jetson Nano with a web monitoring application.
- Developed a RAG chatbot for solar-energy and smart-elevator applications; designed MongoDB/PostgreSQL data structures for logs, alerts and recognition data.

02 SELECTED PROJECTS

ROS2 & Raspberry Pi 5 Autonomous Vehicle

02/2026 - 08/2026

Graduation thesis: integrated lane/sign perception, ROS2 and embedded components on a miniature urban testbed.

Stewart-Platform Ball-Balancing Robot

08/2024 - 05/2025

MATLAB/Simulink modelling; PD / PD-Fuzzy control; integrated camera tracking, LabVIEW and Arduino.

Micromouse Maze-Solving Robot

10/2024 - 12/2024

Main developer: Flood Fill, wall following, motor control, sensors and Bluetooth communication.

03 TECHNICAL SKILLS

Programming & platforms: C/C++, Python, MATLAB; Linux/Ubuntu, Git, Docker.
Robotics & control: ROS/ROS2, CAN bus, PID/PD, PD-Fuzzy; MATLAB/Simulink, LabVIEW.
Embedded & vision: Raspberry Pi 5, Jetson Nano, ESP32/STM32, Arduino; OpenCV, YOLO.
CAD & data/AI: SolidWorks, assemblies, technical drawings; RAG, MongoDB, PostgreSQL.

04 EDUCATION

Ho Chi Minh City University of Technology (HUTECH)

08/2022 - 08/2026

Bachelor in Robotics and Artificial Intelligence - Excellent classification
GPA: 3.74 / 4.0 | Ho Chi Minh City
Main fields: Robotics, Embedded Systems, Automatic Control, Computer Vision, AI, IoT.
Graduation thesis: Autonomous Vehicle System Using ROS2 and Raspberry Pi.
Vietnamese: native | English: B1 - technical reading and communication

05 CERTIFICATIONS, RECOGNITION & RESEARCH

Certifications: Certified SOLIDWORKS Associate (CSWA); Generative AI for Educators with Gemini.
Awards: 1st Prize, Robot Sumo Fight - HUTECH (2023); Top 5, HUTECH Mysterious Maze (2024).
HUTECH student research proceedings: Arduino greenhouse monitoring/control (2023); YOLOv8 traffic-sign recognition (2024); platform ball-balancing robot (2025).